№ 15 $\quad \xi \sim R(\theta, 2\theta) \quad M\xi = \int_\theta^{2\theta} \frac{x\,dx}{\theta} = \frac{3}{2}\theta \qquad \theta > 0$

$f(x,\theta) = \frac{1}{2\theta-\theta}\{(\theta;2\theta)\}$

$M\xi^2 = \int_\theta^{2\theta} \frac{x^2}{\theta}dx = \frac{7\theta^2}{3}$

$f(x,\theta) = \frac{1}{\theta}\{(\theta;2\theta)\}$

$D\xi = M\xi^2 - [M\xi]^2 = \frac{7\theta^2}{3} - \frac{9}{4}\theta^2 = \frac{\theta^2}{12}$

а)
б) $\alpha_1 = \frac{3}{2}\theta \quad \alpha_2 = \frac{7\theta^2}{3} \quad \mu_2 = \frac{\theta^2}{12}$

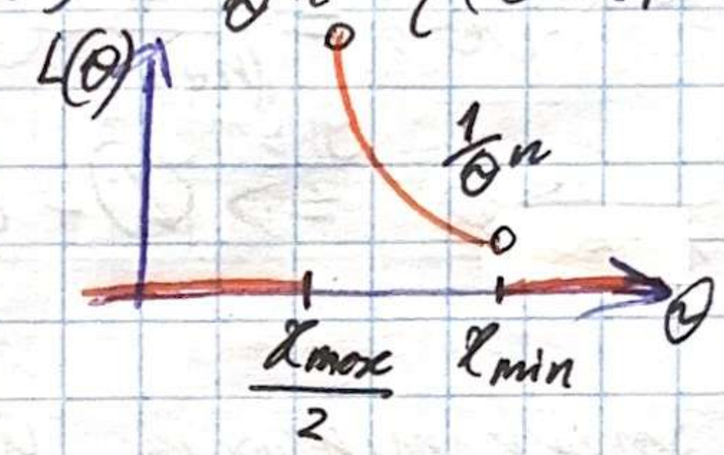
ОММ: $\alpha_1(\theta) = \tilde{\alpha}_1 \quad \frac{3}{2}\theta = \bar{x} \Rightarrow \tilde{\theta}_1 = \frac{2}{3}\bar{x}$ – ОММ.

несм.: $M\tilde{\theta}_1 = \theta$: $M\tilde{\theta}_1 = M\left(\frac{2}{3}\bar{x}\right) = \frac{2}{3}M\xi = \frac{2}{3}\cdot\frac{3}{2}\cdot\theta = \theta$

$\Rightarrow \tilde{\theta}_1$ – несмещ.

сост.: $D\tilde{\theta}_1 = \frac{4}{9n}D\xi = \frac{4\theta^2}{18n} = \frac{\theta^2}{2n} \xrightarrow[n\to\infty]{} 0 \overset{\text{усл. сост.}}{\Rightarrow} \tilde{\theta}_1$ – сост.

ОМП: $L(\theta) = \prod_{i=1}^{n} f(x_i,\theta) \Rightarrow L(\theta) = \frac{1}{\theta^n}\{(\theta < x_i < 2\theta)\}$



$\Rightarrow x_{max} < 2\theta$

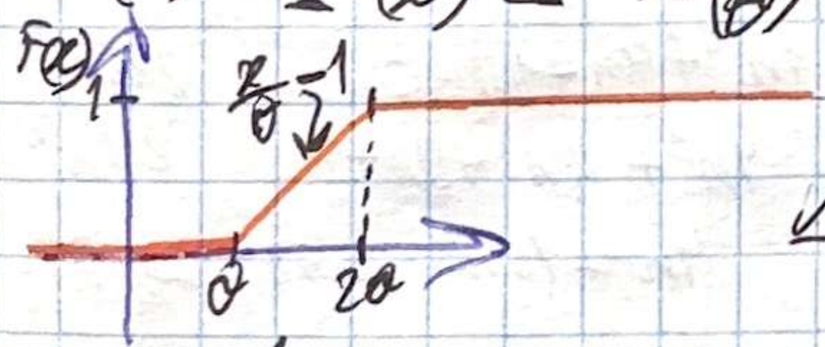
$x_{min} > \theta$

$\Rightarrow \theta \in \left(\frac{x_{max}}{2}; x_{min}\right)$

$\Rightarrow \tilde{\theta}_2 = \frac{x_{max}}{2}$ – ОМП.

несм.: $M\tilde{\theta}_2 = \theta$: $M\tilde{\theta}_2 = M\frac{x_{max}}{2} \qquad x_{max} \sim \underbrace{(F(x))^n}_{\Psi(x)}$

$\varphi(x) = \Psi'(x) = n\left(\frac{x}{\theta}\right)^{n-1}\frac{1}{\theta}\{(\theta;2\theta)\}$



$y = \left(\frac{x}{\theta} - 1\right)$

$\frac{M x_{max}}{2} = \frac{1}{2}\int_\theta^{2\theta} x\, n\left(\frac{x}{\theta}-1\right)^{n-1}\frac{1}{\theta}dx = \frac{n\theta}{2}\int_0^1 (1+y)y^{n-1}dy =$

$= \frac{n\theta}{2}\int_0^1 (y^{n-1} + y^n)dy = \frac{n\theta}{2}\left(\frac{1}{n} + \frac{1}{n+1}\right) = \frac{n\theta}{2}\cdot\frac{(2n+1)}{n(n+1)} = \frac{\theta(2n+1)}{2(n+1)} \Rightarrow$

$\tilde{\theta}_2$ – смещ., исправим: $\tilde{\theta}_2' = \frac{2(n+1)}{(2n+1)}\tilde{\theta}_2 = \frac{(n+1)}{(2n+1)}x_{max} \Rightarrow$

$M\tilde{\theta}_2' = \theta \Rightarrow \tilde{\theta}_2'$ – несмещ.

сост: $M\left[\frac{x_{max}}{2}\right]^2 = \frac{n\theta^2}{4}\int_0^1 (1+y)^2 y^{n-1}\,dy = \frac{n\theta^2}{4}\left(\frac{1}{n}+\frac{2}{n+1}+\frac{1}{n+2}\right) =$

$= \frac{\theta^2}{4}\,\frac{(n+1)(n+2)+2n(n+2)+n(n+1)}{(n+1)(n+2)} = \frac{\theta^2(4n^2+8n+2)}{4(n+1)(n+2)}$

$D\tilde{\theta}_2 = M\tilde{\theta}_2^2 - (M\tilde{\theta}_2)^2 = \frac{\theta^2(4n^2+8n+2)}{4(n+1)(n+2)} - \frac{\theta^2(4n^2+4n+1)}{4(n+1)^2} =$

$= \frac{\theta^2}{4}\left(\frac{(4n^2+8n+2)(n+1)-(4n^2+4n+1)(n+2)}{(n+1)^2(n+2)}\right) = \frac{n\theta^2}{4(n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0$

$\overset{\text{дост. усл.}}{\Longrightarrow} \tilde{\theta}_2$ – сост. ✓

рассм. $\tilde{\theta}_2' = \frac{n+1}{(2n+1)}x_{max} = \frac{2n+2}{2n+1}\,\frac{x_{max}}{2}$

$M[\tilde{\theta}_2']^2 = \left(\frac{2n+2}{2n+1}\right)^2 M\left[\frac{x_{max}}{2}\right]^2 = \frac{(2n+2)^2\theta^2(4n^2+8n+2)}{4(2n+1)^2(n+1)(n+2)}$ ...

Можно проще: $D\tilde{\theta}_2' = \frac{(2n+2)^2}{(2n+1)^2}D\tilde{\theta}_2 = \frac{\cancel{(2n+2)^2}}{(2n+1)^2}\,\frac{n\theta^2}{\cancel{4(n+1)^2}(n+2)} =$

$= \frac{n\theta^2}{(2n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0 \overset{\text{дост. усл.}}{\Longrightarrow} \tilde{\theta}_2'$ – сост. ✓

c) Сравним эффективность исправл. оц.

$\tilde{\theta}_1 = \frac{2}{3}\bar{x}$ $\quad \tilde{\theta}_2' = \frac{n+1}{(2n+1)}x_{max}$

$D\tilde{\theta}_1 = \frac{\theta^2}{27n}$ $\quad D\tilde{\theta}_2' = \frac{n\theta^2}{(2n+1)^2(n+2)}$ $\qquad \frac{1}{27n} \;?\; \frac{n}{(2n+1)^2(n+2)}$

$4n^2-11n-2=0$

$n_{1,2} = \frac{11}{8} \pm \frac{3}{8}\sqrt{17} \Rightarrow$

$n_1 \approx 3$

$n_2 \approx 0$

$(4n^2+4n+1)(n+2)$ $\qquad 27n^2$

$4n^3+12n^2+9n+2$ $\qquad 27n^2$

$4n^3-15n^2+9n+2 > 0$ (пусть)

$n=1$ – графики

$(n-1)(4n^2-11n-2)$

$-\quad +\quad -\quad +$

$n_2 = 0 \quad n_3 = 1 \quad n_1 = 3$

т.к. n – объём выборки

$\Downarrow$ после $n=3$

$D\tilde{\theta}_1 \geq D\tilde{\theta}_2'$

$\Rightarrow \tilde{\theta}_2'$ – эффективнее $\tilde{\theta}_1$

d) Построим точный дов. интерв. для θ (β)

$$f(\vec{x}_n, h) = \frac{x_{max}}{\theta} - 1$$

см. ф.р. $F(x)$

$$P(f<t) = P(x_{max} < \theta t + \theta) = (F(\theta t + \theta))^n = \begin{cases} 0, & t \le 0 \\ t^n, & 0 < t \le 1 \\ 1, & t > 1 \end{cases}$$

Строим интервал:

$$t_1: \ F(t_1) = \frac{1-\beta}{2} \Rightarrow \sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}$$

$$t_2: \ F(t_2) = \frac{1+\beta}{2} \qquad P\left(\sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}\right) = \beta$$

$$\sqrt[n]{\frac{1-\beta}{2}} + 1 < \frac{x_{max}}{\theta} < \sqrt[n]{\frac{1+\beta}{2}} + 1 \Rightarrow \boxed{\frac{x_{max}}{\sqrt[n]{\frac{1+\beta}{2}} + 1} < \theta < \frac{x_{max}}{\sqrt[n]{\frac{1-\beta}{2}} + 1}}$$

$\Rightarrow$ построили точный дов. инт.

e) Построить асимпт. дов. инт для θ $\quad \tilde{\alpha}_1 = \bar{x}$

<u>ОМП:</u> $\tilde{\theta}_1 = \frac{2}{3}\tilde{\alpha}_1 = g(\tilde{\alpha}_1)$ $\qquad \nabla g(\alpha) = \frac{2}{3}$ $\qquad \theta = g(\alpha)$ $\qquad K_{11} = \alpha_{11} - \alpha_1 \alpha_1$

$$\sqrt{n}\,\frac{g(\tilde{\alpha}) - g(\alpha)}{\sigma(\tilde{\alpha})} \xrightarrow{d} N(0,1) \qquad \sigma^2(\alpha) = \nabla^T g(\alpha) K \nabla g(\alpha)$$

$$\sigma^2(\alpha) = \frac{2}{3}(\alpha_2 - \alpha_1^2)\frac{2}{3} = \frac{4}{9}(\alpha_2 - \alpha_1^2)$$

$$\frac{\sqrt{n}(\tilde{\theta}_1 - \theta)}{\frac{2}{3}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} \xrightarrow{d} N(0,1) \qquad u_{\frac{1-\beta}{2}} = t_1, \quad u_{\frac{1+\beta}{2}} = t_2 \Rightarrow P\left(t_1 < \frac{\sqrt{n}(\tilde{\theta}_1 - \theta)}{\frac{2}{3}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} < t_2\right) = \beta$$

$$t_1 \cdot \frac{2\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}}{3 \cdot \sqrt{n}} < \tilde{\theta}_1 - \theta < t_2 \cdot \frac{2 \cdot \sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}}{3 \cdot \sqrt{n}} \Rightarrow \boxed{\tilde{\theta}_1 - \frac{2\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}}{3\sqrt{n}} t_2 <}$$

$$\boxed{< \theta < \tilde{\theta}_1 - \frac{2\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}}{3 \cdot \sqrt{n}} t_1}$$ — построили ас. дов. инт. для ОМП

(Продолжение в TS.ipynb)

№6

$$p(x) = \begin{cases} \frac{\theta-1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases} \qquad \theta > 1 \qquad \overrightarrow{X_n} \quad \beta \text{ выборка}$$

а) ОМП: $L(\theta) = \frac{(\theta-1)^n}{\prod_{i=1}^{n} x_i^{\theta}} = \prod_{i=1}^{n} p(x_i, \theta)$

$$\ln L(\theta) = n \ln(\theta-1) - \theta \sum_{i=1}^{n} \ln x_i \Rightarrow \frac{\partial \ln L(\theta)}{\partial \theta} = \frac{n}{\theta-1} - \sum_{i=1}^{n} \ln x_i = 0$$

$$\Rightarrow \frac{n}{\theta-1} = \sum_{i=1}^{n} \ln x_i \quad \frac{1}{n}\sum_{i=1}^{n} \ln x_i = \frac{1}{\theta-1} \quad \Rightarrow \underset{\text{ОМП}}{\hat{\theta}} = 1 + \frac{1}{\frac{1}{n}\sum_{i=1}^{n} \ln x_i}$$

Проверим на макс.: $\frac{\partial^2 \ln L(\theta)}{\partial \theta^2} = -\frac{n}{(\theta-1)^2} < 0 \quad \forall \theta > 1 \Rightarrow$ лок. макс.

b) Для того, чтобы воспользоваться (Т) об асимп. св-вах ОМП разберём условия:

а) $\Xi = (1; +\infty)$ – открытое ✓

б) сильная рег.: 1) рег. 2) $\frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} dF(x,\theta) = \int_{-\infty}^{+\infty} d\frac{\partial^2 F(x,\theta)}{\partial \theta^2}$

1) рег.: 3) $p(x,\theta) \in C^2$

1.1) $p(x,\theta)$ – непр. дифф. по θ на $(1; +\infty)$ ✓

1.2) $\int_1^{+\infty} \frac{\partial p(x,\theta)}{\partial \theta} dx = \int_1^{+\infty} \frac{\ln x - \theta \ln x + 1}{x^{\theta}} dx = \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty} + \int_1^{+\infty} \ln x \, d\left(\frac{x^{1-\theta}}{1-\theta}\right) =$

$= \frac{1}{\theta-1} - \frac{1}{\theta-1} = 0$ ✓

1.3) $I(\theta) = M\left[\left(\frac{\partial \ln p(x,\theta)}{\partial \theta}\right)^2\right] = \int_1^{+\infty} \frac{\theta-1}{x^{\theta}} \left(\frac{1}{\theta-1} - \ln x\right)^2 dx =$

$$= \int_1^{+\infty} \left(\frac{1}{x^{\theta}(\theta-1)} - 2\frac{\cancel{(\theta-1)}\ln x}{x^{\theta}\cancel{(\theta-1)}} + \frac{(\theta-1)\ln^2 x}{x^{\theta}}\right) dx = \frac{1}{(\theta-1)^2} - 2\underbrace{\int_1^{+\infty} \frac{\ln x}{x^{\theta}} dx}_{I_1} +$$

$$+ \underbrace{\int_1^{+\infty} \frac{(\theta-1)\ln^2 x}{x^{\theta}} dx}_{I_2} =$$

$$I_1 = -2\int_1^{+\infty} \ln x \, d\left(-\frac{1}{(\theta-1)x^{\theta-1}}\right) = -2 \cancel{\frac{\ln x \, x^{1-\theta}}{\theta-1}\Big|_1^{+\infty}} + 2\int_1^{+\infty} \frac{x^{-\theta}}{1-\theta}dx =$$

$$= 2 \cdot \frac{1}{(1-\theta)^2}$$

$$I_2 = \int_1^{+\infty} \frac{(\theta-1)}{x^\theta} \ln^2 x \, dx = \int_1^{+\infty} \ln^2 x \, d(-x^{1-\theta}) = \cancel{-x^{(1-\theta)} \ln^2 x \Big|_1^{+\infty}} + 2\int_1^{+\infty} \frac{x^{1-\theta} \ln x}{x} dx =$$

$$= 2\int_1^{+\infty} x^{-\theta} \ln x \, dx = \cancel{\frac{2x^{1-\theta}}{1-\theta} \ln x \Big|_1^{+\infty}} - 2\int_1^{+\infty} \frac{x^{-\theta}}{1-\theta} dx = -2 \frac{1}{(1-\theta)^2}$$

$$\Rightarrow \ominus \frac{1}{(\theta-1)^2} + \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = \frac{1}{(\theta-1)^2}$$ – непр. на $(1; +\infty)$

$I(\theta) > 0$ на

$\Rightarrow$ модель регулярна

2) $$\frac{\partial^2 p(x,\theta)}{\partial\theta^2} = \frac{-2\ln x + \ln^2 x\,(\theta-1)}{x^\theta}$$

$$\frac{\partial^2}{\partial\theta^2} \int_1^{+\infty} p(x,\theta)dx = 0 \qquad \int_1^{+\infty} \frac{\partial^2 p(x,\theta)}{\partial\theta^2} dx = \int_1^{+\infty} \Big(\underbrace{-\frac{2\ln x}{x^\theta}}_{I_1} + \underbrace{\frac{\ln^2 x(\theta-1)}{x^\theta}}_{I_2}\Big) dx =$$

$$= \frac{2}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = 0$$

3) $p(x,\theta)$ 2жды непр. дифф. по θ на $(1; +\infty)$

$\Rightarrow$ модель сильно регулярна $\Rightarrow$ восп.

$\Rightarrow$ ОМП – сост., ас. несм., ас. норм., ас. эфф.

медианы: $p = 0{,}5$ $\quad F(x_p) = p$ $\quad F(x_{0,5}) = 0{,}5$

Найдём ф.р.: $$\int_1^x \frac{\theta-1}{t^\theta} dt = -\frac{1}{x^{\theta-1}} + 1 = \frac{1}{2}$$

$$\Rightarrow x_{0,5} = 2^{\frac{1}{\theta-1}} \Rightarrow g(\theta) = 2^{\frac{1}{\theta-1}} \qquad g(\theta) = 2^{\frac{1}{\theta-1}}$$

$$\nabla g(\theta) = 2^{\frac{1}{\theta-1}} \cdot \ln 2 \cdot -\frac{1}{(\theta-1)^2}$$

$$I^{-1}(\theta) = (\theta-1)^2 \qquad \Rightarrow \sigma^2(\theta) = \nabla^T g(\theta) I^{-1}(\theta) \nabla g(\theta)$$

$$\sigma^2(\theta) = \left(2^{\frac{1}{\theta-1}} \ln 2 \left(-\frac{1}{(\theta-1)^2}\right)\right)^2 \cdot (\theta-1)^2$$

$$\Rightarrow \sqrt{n}\,\frac{\left(2^{\frac{1}{\tilde\theta-1}} - 2^{\frac{1}{\theta-1}}\right)}{2^{\frac{1}{\tilde\theta-1}} \cdot \ln 2 \cdot \frac{1}{(\tilde\theta-1)}} \rightsquigarrow N(0,1)$$

$t_1 = u_{\frac{1-\beta}{2}}$

$t_2 = u_{\frac{1+\beta}{2}}$

$$P\left(t_1 < \frac{\sqrt{n}\,(\tilde\theta-1)\cdot\left(2^{\frac{1}{\tilde\theta-1}} - \boxed{2^{\frac{1}{\theta-1}}}\right)}{2^{\frac{1}{\tilde\theta-1}} \cdot \ln 2} < t_2\right) = \beta$$

$x_{0,5}$

$$\Rightarrow \frac{2^{\frac{1}{\tilde\theta-1}} \cdot \ln 2}{\sqrt{n}\,(\tilde\theta-1)}\, t_1 < 2^{\frac{1}{\tilde\theta-1}} - x_{0,5} < \frac{2^{\frac{1}{\tilde\theta-1}} \cdot \ln 2}{\sqrt{n}\,(\tilde\theta-1)}\, t_2$$

$$\Rightarrow \boxed{2^{\frac{1}{\tilde\theta-1}} - \frac{2^{\frac{1}{\tilde\theta-1}} \cdot \ln 2}{\sqrt{n}\,(\tilde\theta-1)}\, t_2 < x_{0,5} < 2^{\frac{1}{\tilde\theta-1}} - \frac{2^{\frac{1}{\tilde\theta-1}} \cdot \ln 2}{\sqrt{n}\,(\tilde\theta-1)}\, t_1}$$

– построили асимпт. дов. инт. для медианы по ОМП

c) $\sqrt{n}\,\dfrac{\tilde\theta-\theta}{\sigma(\tilde\theta)} \rightsquigarrow N(0,1)$

$g(\theta) = \theta \Rightarrow \nabla g(\theta) = 1$

$\sigma^2(\tilde\theta) = 1 \cdot (\theta-1)^2 \cdot 1$

$\sqrt{n}\,\dfrac{\tilde\theta-\theta}{(\theta-1)} \rightsquigarrow N(0,1)$

$\tilde\theta = 1 + \dfrac{1}{\frac{1}{n}\sum_{i=1}^{n} \ln x_i}$

$\sqrt{n}\,\frac{1}{n}\sum_{i=1}^{n} \ln x_i\,(\tilde\theta-\theta) \rightsquigarrow N(0,1)$

$\Rightarrow \tilde\theta - 1 = \dfrac{1}{\frac{1}{n}\sum_{i=1}^{n} \ln x_i} \qquad \dfrac{1}{\tilde\theta-1} = \dfrac{1}{n}\sum_{i=1}^{n} \ln x_i$

$t_1 = u_{\frac{1-\beta}{2}}$

$t_2 = u_{\frac{1+\beta}{2}}$

$$P\left(t_1 < \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \ln x_i\,(\tilde\theta-\theta) < t_2\right) = \beta$$

$$\frac{t_1}{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \ln x_i} < \tilde\theta - \theta < \frac{t_2}{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \ln x_i}$$

$$\Rightarrow \boxed{\tilde\theta - \frac{t_2}{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \ln x_i} < \theta < \tilde\theta - \frac{t_1}{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \ln x_i}}$$ – построили асимпт. дов. инт. для θ по ОМП

(Продолжение в Тб.ipynb)